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Three-dimensional steam generating system for performing a steam cycle within a laundry appliance

Abstract

A laundry appliance includes a blower that delivers process air through an airflow path. A rotating drum defines a processing chamber. The processing chamber is part of the airflow path. A steam generating system disposes steam into the processing chamber. A plurality of fluid nozzles direct the steam into the processing chamber according to an operating pattern. The plurality of fluid nozzles include a first nozzle that is positioned within an upper portion of the processing chamber and a second nozzle that is positioned in a lower portion of the processing chamber.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to and the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 63/189,289, filed on May 17, 2021, entitled THREE-DIMENSIONAL STEAM GENERATING SYSTEM FOR PERFORMING A STEAM CYCLE, the entire disclosure of which is hereby incorporated herein by reference.

FIELD OF THE DEVICE

(1) The device is generally in the field of laundry appliances, and more specifically, a laundry appliance including a plurality of fluid-directing nozzles that are operable to perform a variety of steam functions that can be adjusted based upon various parameters.

SUMMARY OF THE DEVICE

(2) According to one aspect of the present disclosure, a laundry appliance includes a blower that delivers process air through an airflow path. A rotating drum defines a processing chamber. The processing chamber is part of the airflow path. A steam generating system disposes steam into the processing chamber. A plurality of fluid nozzles direct the steam into the processing chamber according to an operating pattern. The plurality of fluid nozzles include a first nozzle that is positioned within an upper portion of the processing chamber and a second nozzle that is positioned in a lower portion of the processing chamber.

(3) According to another aspect of the present disclosure, a laundry appliance includes a cabinet having an access aperture defined within a front panel of the cabinet. A drum rotates within the cabinet. The access aperture provides selective access to a processing chamber defined within the drum. A steam generating system delivers steam to the processing chamber. A first nozzle is coupled to the front panel and above the access aperture. A second nozzle is positioned within a rear panel of the processing chamber. A steam generator delivers steam to the first nozzle and the second nozzle according to an operating pattern. The operating pattern is determined by a plurality of sensors that are positioned in communication with the processing chamber. The steam generator generates steam in the absence of a dedicated heating element.

(4) According to yet another aspect of the present disclosure, a laundry appliance includes a cabinet having an access aperture. A drum rotates within the cabinet to process a load of articles. The access aperture provides selective access to a processing chamber defined within the drum. A steam generating system delivers steam to the processing chamber. A first nozzle is coupled to a front panel and above the access aperture. A second nozzle is positioned within a rear panel of the processing chamber. The steam generating system delivers steam to the first nozzle and the second nozzle according to an operating pattern. The operating pattern is determined by a plurality of sensors that are positioned in communication with the processing chamber. The first nozzle is oriented to direct a first jet of fluid in a downward direction toward a rear panel of the processing chamber and the second nozzle is positioned within the rear panel and is oriented to direct a second jet of fluid in an upward direction toward the access aperture. The first and second jets of fluid define a three-dimensional pattern of steam that engages the load of articles from above and below, respectively.

(5) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

- (2) FIG. 1 is a schematic cross-sectional view of a laundry appliance that incorporates an aspect of the steam generating system;
- (3) FIG. 2 is a schematic perspective view of a steam generating system for a laundry appliance;
- (4) FIG. 3 is an elevational view of an aspect of a user interface that is incorporated within a laundry appliance for operating a steam cycle;
- (5) FIG. 4 is a front perspective view of a laundry appliance that incorporates an aspect of a steam generating system;
- (6) FIG. 5 is a perspective view of the laundry appliance of FIG. 4 showing a door in the open position and a position of one of the fluid nozzles of a plurality of fluid nozzles for the steam generating system;
- (7) FIG. 6 is a front perspective view of a laundry appliance showing steam being generated within the processing chamber;
- (8) FIG. 7 is a schematic flow diagram illustrating an aspect of the selection program for a user interface for initiating a steam function of a laundry appliance;
- (9) FIG. 8 is a schematic flow diagram illustrating an algorithm for estimating a size of a load being processed within a laundry appliance;
- (10) FIG. 9 is a schematic flow diagram illustrating a process for calculating load size and activating a steam function for a laundry appliance;
- (11) FIG. 10 is a schematic flow diagram illustrating a plurality of steam sequences that are operated based upon a size of a load being processed within the appliance;
- (12) FIG. 11 is a schematic flow diagram illustrating a plurality of steam sequences that are operated based upon a size of a load being processed within the appliance;
- (13) FIG. 12 is a schematic flow diagram illustrating a plurality of steam sequences that are operated based upon a size of a load being processed within the appliance;
- (14) FIG. 13 is a schematic flow diagram illustrating operation of a conductivity sensor for managing an amount of steam generated within the processing chamber;
- (15) FIG. 14 is a schematic flow diagram illustrating a process of performing a steam function using the plurality of fluid nozzles in using the conductivity strip for assessing the amount of steam generated within the processing chamber; and
- (16) FIG. 15 is a schematic flow diagram illustrating a method for generating steam with a laundry appliance.
- (17) The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles described herein.

DETAILED DESCRIPTION

(18) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a steam generating system for a laundry appliance that operates according to a sequence that is adjustable based upon various parameters. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

Further, like numerals in the description and drawings represent like elements.

(19) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the disclosure as oriented in FIG. 1. Unless stated otherwise, the term “front” shall refer to the surface of the element closer to an

intended viewer, and the term “rear” shall refer to the surface of the element further from the intended viewer. However, it is to be understood that the disclosure may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(20) The terms “including,” “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises a . . .” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(21) Referring now to FIGS. **1-8**, reference numeral **10** generally refers to a steam generating system for a laundry appliance **12** that generates steam **14** within a processing chamber **16** of the appliance **12**, typically within a rotating drum **18**. A steam generating system **10** includes a plurality of fluid nozzles **20** that are used to generate steam **14** within the processing chamber **16**. According to various aspects of the device, the appliance **12** includes a blower **22** that delivers process air **24** through an airflow path **26**. The rotating drum **18** defines the processing chamber **16**. The processing chamber **16** is included within the airflow path **26** for receiving process air **24** that dehumidifies articles **28** contained within the processing chamber **16**. The steam generating system **10** operates to dispose steam **14** within the processing chamber **16**. The plurality of fluid nozzles **20** operate to direct the steam **14** into and through the processing chamber **16** according to a predetermined operating pattern **30**. According to various aspects of the device, the predetermined operating pattern **30** can be based upon various factors. These factors can include, but are not limited to, the type of article **28** or fabric that is being processed within the processing chamber **16**, an amount of articles **28** that are being processed within the processing chamber **16**, combinations thereof and other factors.

(22) Typically, the predetermined operating pattern **30** can vary depending upon various parameters relating to the selected operating cycle **32** and the articles **28** being processed within the processing chamber **16**. These predetermined operating patterns **30** typically include selective operation of the steam generating system **10**. This selective operation is in the form of activations **34** and deactivations **36** of the plurality of fluid nozzles **20**. These activations **34** and deactivations **36** can be sequential, alternating, combinations thereof and other operating patterns **30** of activations **34** and deactivations **36** of one or more of the plurality of fluid nozzles **20**. The activations **34** and deactivations **36** of the operating pattern **30** can be modified or adjusted based upon the measurements taken by one or more sensors, and typically a plurality of sensors. The sensors are configured to take measurement of the process air **24**, the conditions within the processing chamber **16** and the conditions of the articles **28** being processed. Using these measurements, the controller **170** can assess whether any of the activations **34** or the deactivations **36** of the operating pattern **30** need to be extended, suspended or otherwise modified.

(23) According to various aspects of the device, the steam generating system **10** includes a steam generator that generates steam **14** within the processing chamber **16** without the use of a dedicated heat source within the steam generating system **10**. Accordingly, the steam generating system **10** generates steam **14** having a temperature that is similar to that of the air temperature of the process air **24** within the processing chamber **16**. In this manner, fluid **82** is delivered to the fluid nozzles **20** and is sprayed into the processing chamber **16** in the form of a fine mist. Various differences in humidity levels, dew points, and other environmental factors between areas outside of the processing chamber **16** and the areas inside of the processing chamber **16** cooperate to generate steam **14** within the processing chamber **16**. Various air handlers can be used to move the steam **14**

that is generated within the processing chamber **16** around and through the articles **28** to be processed in the laundry appliance **12**.

(24) Referring now to FIGS. **1-6**, the plurality of fluid nozzles **20** can include a front fluid nozzle **50**, also referred to herein as the first nozzle, that is positioned to generate steam **14** within an upper portion **52** of the processing chamber **16**, and a rear fluid nozzle **54** that is positioned to generate steam **14** within a lower portion **56** of a processing chamber **16**. The front fluid nozzle **50** can be positioned near or within a front panel **58** of the appliance **12** so that the fluid nozzle **20** is directed to inject or otherwise dispose a first jet **70** of fluid **82** in a generally downward direction into the access aperture **122** and toward a rear panel **60** of the rotating drum **18**. The rear fluid nozzle **54**, also referred to herein as the second nozzle, is typically positioned within the rear panel **60** for the rotating drum **18** that is generally stationary as a perimetrical wall **62** of the rotating drum **18** rotates about a rotational axis **64**. The rear fluid nozzle **54** is adapted to direct a second jet **72** of fluid **82** in a generally upward direction and toward the access aperture **122**. When both the front and rear fluid nozzles **50**, **54** are activated, steam **14** is generated within a front portion **66** and the upper portion **52** of the processing chamber **16** and a rear portion **68** and the lower portion **56** of the processing chamber **16**. In this manner, the steam **14** generated by a plurality of fluid nozzles **20** can be moved throughout the entire processing chamber **16** for penetrating within, around and through the articles **28** being processed within the rotating drum **18**. This configuration of the steam generating system **10** also allows the articles **28** to tumble through areas of steam **14** that are generated in the processing chamber **16**.

(25) Referring again to FIGS. **1** and **2**, the steam generating system **10** includes a fluid delivery manifold **80** that directs a fluid **82** to the front and rear fluid nozzles **50**, **54**. Various valves **84** are attached to the fluid delivery manifold **80** and/or the front and rear fluid nozzles **50**, **54** for selectively operating the front and rear fluid nozzles **50**, **54** (first and second nozzles). This selective operation of the front and rear fluid nozzles **50**, **54** is used to perform the various activations **34** and deactivations **36** of the steam generating system **10** and the plurality of fluid nozzles **20**. The front and rear fluid nozzles **50**, **54** of the steam generating system **10** operate to define a three-dimensional pattern **74** of steam **14** that engages the articles **28** within the processing chamber **16** from the front and from the rear as well as from above and below, as described more fully herein.

(26) Referring now to FIG. **3**, a user interface **100** for the appliance **12** can include dedicated controls for operating the steam generating system **10** during performance of any one or more operating cycles **32** of the appliance **12**. Certain operating cycles **32** can include a built-in steam function **102** that automatically operates within a portion of the operating cycle **32**. It is also contemplated that various steam functions **102** can be specifically selected utilizing the user interface **100**. These specific selections can activate or deactivate a steam function **102**, increase or decrease the amount of steam **14** generated during a processing chamber **16**, set various parameters regarding the amount of steam **14** to be generated, and other similar factors that are described more fully herein.

(27) Referring now to FIGS. **4-6**, the front fluid nozzle **50** is typically positioned at a 12-o'clock position of a rotating drum **18** and near a top portion **120** of the access aperture **122** of the rotating drum **18**. The front fluid nozzle **50** may be attached to the outer cabinet **124** or attached to a structural member of the appliance **12** near the outer cabinet **124**. In this manner, the front fluid nozzle **50** is stationary as the drum **18** rotates about the rotational axis **64**. The front fluid nozzle **50** directs the fluid **82** through the access aperture **122** and into the processing chamber **16** so that steam **14** can be generated within the processing chamber **16**. In various aspects of the device, the door panel **126** for the appliance **12** can include a window so that a user can view the generation of steam **14** within the processing chamber **16**.

(28) Referring now to FIGS. **7-9**, the steam generating system **10** can include an automatic activation sequence **140** that can be used to determine the appropriate amount of steam **14** to be

generated within the processing chamber **16**. While various sensors and data points can be utilized for estimating the amount of articles **28** being processed or the type of articles **28** being processed, an exemplary aspect of the device includes rear and front temperature probes **142**, **144** that are positioned at an upstream location **146** of the processing chamber **16** and a downstream location **148** of the processing chamber **16**, respectively. As exemplified in FIG. **1**, the front temperature probe **144** is positioned at the downstream location **148** so that process air **24** leaving the processing chamber **16** can be monitored by the front temperature probe **144** for determining temperature of the process air **24** leaving the processing chamber **16**. The rear temperature probe **142** monitors the temperature of the process air **24** entering into the processing chamber **16**, and is typically positioned behind the rotating drum **18** within a portion of the airflow path **26** proximate a back wall **150** of the appliance **12**.

(29) The rear and front temperature probes **142**, **144** monitor a change or “A” of the temperature for the process air **24** as it moves through the processing chamber **16** and engages the various articles **28** to dehumidify the articles **28** during the laundry cycle. The change in temperature, or the temperature difference, of the process air **24** can have a predictable temperature change depending upon the amount of articles **28** being processed within the processing chamber **16**. Utilizing this temperature difference between the temperature of the process air **24** entering the processing chamber **16** and the temperature of the process air **24** leaving the processing chamber **16**, the amount of articles **28** being processed can be estimated and categorized between a small load, medium load and large load. It is also contemplated that the load sizes can be sorted into fewer categories, such as large load and small load. Greater numbers of categories can also be used depending upon the sensitivity of the rear and front temperature probes **142**, **144** and the desired accuracy of the various steam functions **102**.

(30) Referring again to FIGS. **8** and **9**, when the size of the load is estimated, the amount of steam **14** to be generated within the processing chamber **16** can be increased for larger loads and decreased for smaller loads. The amount of steam **14** to be generated is intended to be an appropriate amount for refreshing or de-wrinkling clothing without causing an undesirable accumulation of moisture within the processing chamber **16** that may cause the articles **28** to become over saturated with fluid **82**.

(31) As exemplified in FIGS. **8** and **9**, the amount of moisture to be generated within the processing chamber **16** is a value that can be calculated by a controller **170** for the appliance **12**. The performance of generating steam **14** within the processing chamber **16** is undertaken by the various activations **34** and deactivations **36** of the front and rear fluid nozzles **50**, **54**. By way of example, and not limitation, a greater amount of steam **14** that is to be generated within the processing chamber **16** can result in a greater amount of activations **34** or longer periods of activation **34** between the front and rear fluid nozzles **50**, **54**. Where lesser amounts of steam **14** are intended to be generated, there may be a greater degree of deactivations **36** or longer period of deactivation **36** between the front and rear fluid nozzles **50**, **54** for the steam generating systems **10**. Stated another way, more activations **34** and/or longer activations **34** generally result in a greater amount of steam **14** that is generated within the processing chamber **16**.

(32) Referring now to FIGS. **10-12**, depending upon the amount of articles **28** that are estimated to be present within the processing chamber **16**, the operating pattern **30** that includes various configurations of activations **34** and deactivations **36** of the front and rear fluid nozzles **50**, **54** can vary. As exemplified in FIG. **10**, three separate small-load configurations **190** for the front and rear fluid nozzles **50**, **54** are presented in an exemplary and non-limiting fashion. The differences in the operating pattern **30** of activations **34** and deactivations **36** can vary depending upon the length of the activation **34** or deactivation **36**, the operating pattern **30** of activations **34** and deactivations **36**, and the alternating and simultaneous activation **34** of the front and rear fluid nozzles **50**, **54**. Similarly, as exemplified in FIG. **11**, three separate large-load configurations **192** of the steam generating system **10** are presented in an exemplary and non-limiting fashion.

(33) As exemplified in FIGS. **10** and **11**, the various activations **34** and deactivations **36** of the front fluid nozzle **50** and rear fluid nozzle **54** tends to be longer in the large-load configurations **192** so that greater amounts of steam **14** can be generated within the processing chamber **16**. Additionally, the operating pattern **30** of activations **34** and deactivations **36** of the front and rear fluid nozzles **50**, **54** can vary between simultaneous activation, sequential activation, prolonged activation or deactivation, and combinations of these variations between the activations **34** and deactivations **36** of the plurality of fluid nozzles **20**.

(34) As exemplified in FIG. **12**, the difference between the small-load configurations **190** and large-load configuration **192** for the steam generating system **10** can be in the form of a number of cycles performed in a recurring fashion during a particular drying cycle. As shown in FIG. **12**, the small-load configuration **190** is repeated ten times where the large-load configuration **192** is repeated 20 times.

(35) As exemplified in FIGS. **8-12**, the estimation of load size using the rear and front temperature probes **142**, **144** can change the configuration of the steam cycle so that more steam **14** is generated when larger loads of articles **28** are present. It is also contemplated that the estimation of load size performed by the rear and front temperature probes **142**, **144** and the controller **170** can extend the time of the cycle so that the additional activations **34** within a large-load configuration **192** can be performed to completion. Alternatively, where a smaller load of articles **28** is being processed, a particular laundry cycle may be shortened after the requisite number of activations **34** and deactivations **36** of the front and rear fluid nozzles **50**, **54** has been performed to completion.

(36) According to various aspects of the device, the rear and front temperature probes **142**, **144** can also be utilized for determining a garment type or fabric type of the articles **28** that are being processed within the processing chamber **16**. Typically, users of laundry appliances **12** tend to group articles **28** to be processed within particular categories. Such categories can include delicates, whites (typically cotton), darks, sheets, linens and other similar fabric-type categories. Article-type categories can include bulky items, non-clothing items (such as shoes), and others. The user interface **100** for the laundry appliance **12** can be configured to provide a selection of garment type. It is also contemplated that the various sensors can be used to estimate a garment type that is being processed. These estimations can be automatically selected or can be part of a confirmation sequence used when selecting a laundry cycle. The confirmation can be related to the particular fabric type or article **28** being processed. The various steam functions **102** presented herein for large and small loads of articles **28** can also vary depending upon the fabric type. Delicates may require lesser amounts of steam **14** where cottons may require larger amounts of steam **14**. These various parameters can be adjusted based upon garment type, load size, selected dryness level and other similar parameters that may be hardwired or programmed within the controller **170** for the appliance **12** or selected by a user, or both.

(37) Referring now to FIGS. **13** and **14**, various mechanisms can be utilized for assessing an amount of steam **14** that has been generated within a laundry appliance **12**. By way of example, and not limitation, a conductivity sensor **210**, in the form of a conductivity strip, can be positioned within the processing chamber **16**. This conductivity sensor **210** can be attached to an inner surface **212** of the drum **18**. The amount of humidity within the processing chamber **16**, in the form of steam **14**, can be sensed by the conductivity sensor **210** for estimating an amount of steam **14** contained within the environment of the processing chamber **16**. When an appropriate amount of steam **14** has been achieved within the processing chamber **16**, the conductivity sensor **210** senses this amount and communicates with a controller **170** to initiate a deactivation **36** of one or both of the front and rear fluid nozzles **50**, **54**. The conductivity sensor **210** can also communicate with the controller **170** to modify the activations **34** and deactivations **36** to provide more or less steam **14** within the processing chamber **16**.

(38) By way of example, and not limitation, over the course of a drying cycle, the amount of steam **14** required within the processing chamber **16** may change over time. As these steam generating

needs change, the conductivity sensor **210** can adjust the activations **34** and deactivations **36** of front and rear fluid nozzles **50, 54** to provide a corresponding increase or decrease in the amount of steam **14** generated therein.

(39) Referring now to FIG. **14**, it is contemplated that the conductivity sensor **210** is used to measure a moisture level in relation to a maximum predefined moisture level. Typically, the conductivity sensor **210** is disposed on an inner surface **212** of the drum **18** and measures an amount of moisture or a humidity level of process air **24** within the processing chamber **16**. In this manner, when the amount of steam **14** generated within the processing chamber **16** reaches a predefined maximum amount of moisture, the conductivity sensor **210** can communicate with the front and rear fluid nozzles **50, 54**, via the controller **170**, to initiate a deactivation **36** of the front and rear fluid nozzles **50, 54**. If the amount of moisture within the processing chamber **16** dips below the predefined maximum amount or below a predefined minimum amount, the conductivity sensor **210** can measure this decrease in moisture and communicate with the controller **170** to initiate an activation **34** of the front and rear fluid nozzles **50, 54** to provide additional amounts of steam **14**, as needed.

(40) Having described various aspects of the steam generating system **10**, a method **400** is disclosed for generating steam **14** within a processing chamber **16**. The method **400** includes selecting a drying function, steam function **102** or other operating cycle **32** (step **402**). As discussed herein, the process of selecting an operating function is typically initiated by a user via a user interface **100**. In certain aspects of the device, it is contemplated that a drying appliance **12** can automatically select certain operating cycles **32** based upon various sensor readings of the articles **28** being processed within the drum **18**. According to the method **400**, the operating cycle **32** is activated (step **404**). When activated, a load size of the articles **28** being processed is estimated (step **406**). The process of estimating load size can be initiated during the typical performance of a particular operating cycle **32**. It is also contemplated that the estimating function can be performed during a dedicated estimation routine that is performed before the selected operating cycle **32** is initiated. Once the load size is estimated, a steam function **102** is initiated based upon the load size (step **408**). During performance of the steam function **102** and the operating cycle **32**, an amount of steam **14** generated within the processing chamber **16** is monitored (step **410**). As discussed herein, this monitoring function is typically performed by a conductivity sensor **210** that is positioned within the processing chamber **16**. According to the method **400**, an amount of steam **14** is maintained within a preferred steam or moisture range that is below a maximum predetermined amount and above a minimum predetermined amount of moisture (step **412**). In this manner, the conductivity sensor **210** communicates with the controller **170** to modify the activations **34** and deactivations **36** of the plurality of fluid nozzles **20** to adjust an amount of steam **14** generated within the processing chamber **16**. The operating cycle **32** is then completed (step **414**).

(41) According to various aspects of the device, the steam generating system **10** is used to provide an automated steam function **102** that can be adjusted on the fly and in real time during performance of various laundry cycles. The steam generating system **10** generates steam **14** within the processing chamber **16** and provides a desired amount of moisture within the processing chamber **16** for performing various refreshing and dewrinkling type steam functions **102** upon articles **28** contained within the processing chamber **16**.

(42) As discussed herein, the steam generating system **10** generates steam **14** within the processing chamber **16** without the aid of a dedicated heating element within the steam generating system **10**. It is contemplated that the various algorithms and steam generating cycles discussed herein can be utilized with steam generating systems **10** that do utilize a heating element for generating steam **14** and injecting jets of steam **14** into the processing chamber **16**. It is also contemplated that a combination of heated steam generating elements and non-heated fluid nozzles **20** can be used in combination for generating various amounts of steam **14** within the processing chamber **16**.

(43) The plurality of fluid nozzles **20** disclosed herein can include the front fluid nozzle **50** and the

rear fluid nozzle **54**. It is also contemplated that additional fluid nozzles **20** can be incorporated within the drum **18** for providing other configurations of a three-dimensional steam generating function of the steam generating system **10**. Where additional fluid nozzles **20** are provided in a steam generating system **10**, the configuration of activations **34** and deactivations **36** of the plurality of fluid nozzles **20** can be adjusted to provide the desired amounts of steam **14** and configurations of steam **14** within the processing chamber **16**. The configurations of steam **14** provided by the plurality of fluid nozzles **20** are intended to provide steam **14** throughout the processing chamber **16** so that the articles **28** receive steam **14** from below via the rear fluid nozzles **54** and also tumbles through a generated section of steam **14** within an upper portion **52** of the processing chamber **16**. This combination of fluid nozzles **20** is configured to provide sufficient steam **14** to allow steam **14** to move within, around and through various articles **28** being processed.

(44) According to another aspect of the present disclosure, a laundry appliance includes a blower that delivers process air through an airflow path. A rotating drum defines a processing chamber. The processing chamber is part of the airflow path. A steam generating system disposes steam into the processing chamber. A plurality of fluid nozzles direct the steam into the processing chamber according to an operating pattern. The plurality of fluid nozzles include a first nozzle that is positioned within an upper portion of the processing chamber and a second nozzle that is positioned in a lower portion of the processing chamber.

(45) According to another aspect, the operating pattern is based upon a fabric type that is being processed in the processing chamber.

(46) According to yet another aspect, the operating pattern is based upon an amount of articles that are being processed in the processing chamber.

(47) According to another aspect of the present disclosure, the operating pattern includes activations and deactivations of the plurality of fluid nozzles.

(48) According to another aspect, the rotating drum operates within the cabinet and includes an access aperture that provides selective access to the processing chamber. The first nozzle is positioned above the access aperture and is oriented to direct a first jet of fluid in a downward direction toward a rear panel of the processing chamber.

(49) According to yet another aspect, the second nozzle is positioned within the rear panel and is oriented to direct a second jet of fluid in an upward direction toward the access aperture.

(50) According to another aspect of the present disclosure, a steam generator generates steam having a temperature that is similar to an air temperature of the process air within the processing chamber. The steam generator delivers fluid to the first nozzle and the second nozzle according to the operating pattern.

(51) According to another aspect, the operating pattern includes selective activation of valves for the first nozzle and the second nozzle, respectively. The steam generator includes the first nozzle and the second nozzle.

(52) According to yet another aspect, a conductivity sensor is in communication with the steam generator and the valves. The conductivity sensor measures a humidity level within the processing chamber. The operating pattern and the selective activation of the valves are operated at least partially according to the humidity level within the processing chamber as measured by the conductivity sensor.

(53) According to another aspect of the present disclosure, a rear temperature probe is positioned within the airflow path and proximate the rear panel of the processing chamber. The rear temperature probe measures an air temperature of the process air entering the processing chamber. A front temperature probe is positioned within the airflow path and proximate a front panel of the processing chamber. The front temperature probe measures the air temperature of the process air exiting the processing chamber. The rear and front temperature probes measure a temperature change of the process air. The operating pattern and the selective activation of the valves are

operated at least partially according to the temperature change of the process air.

(54) According to another aspect, a laundry appliance includes a cabinet having an access aperture defined within a front panel of the cabinet. A drum rotates within the cabinet. The access aperture provides selective access to a processing chamber defined within the drum. A steam generating system delivers steam to the processing chamber. A first nozzle is coupled to the front panel and above the access aperture. A second nozzle is positioned within a rear panel of the processing chamber. A steam generator delivers steam to the first nozzle and the second nozzle according to an operating pattern. The operating pattern is determined by a plurality of sensors that are positioned in communication with the processing chamber. The steam generator generates steam in the absence of a dedicated heating element.

(55) According to yet another aspect, the operating pattern includes selective operation of the first and second nozzles.

(56) According to another aspect of the present disclosure, the first nozzle is positioned within an upper portion of the processing chamber and a second nozzle is positioned in a lower portion of the processing chamber. The second nozzle is positioned within the rear panel and is oriented to direct steam in an upward direction toward the access aperture.

(57) According to another aspect, the steam generating system includes a steam generator that generates steam having a temperature that is similar to an air temperature of process air within the processing chamber. The steam generator delivers steam to the first nozzle and the second nozzle according to the operating pattern.

(58) According to yet another aspect, the steam generating system includes valves that are positioned between the steam generator and the first and second nozzles. The operating pattern includes selective activation of the valves for the first nozzle and the second nozzle, respectively.

(59) According to another aspect of the present disclosure, a conductivity sensor is in communication with the steam generator and the valves. The conductivity sensor is disposed on a surface of the drum and measures a humidity level of process air within the processing chamber. The operating pattern and the selective activation of the valves are operated at least partially according to the humidity level, as measured by the conductivity sensor.

(60) According to another aspect, a laundry appliance includes a cabinet having an access aperture. A drum rotates within the cabinet to process a load of articles. The access aperture provides selective access to a processing chamber defined within the drum. A steam generating system delivers steam to the processing chamber. A first nozzle is coupled to a front panel and above the access aperture. A second nozzle is positioned within a rear panel of the processing chamber. The steam generating system delivers steam to the first nozzle and the second nozzle according to an operating pattern. The operating pattern is determined by a plurality of sensors that are positioned in communication with the processing chamber. The first nozzle is oriented to direct a first jet of fluid in a downward direction toward a rear panel of the processing chamber and the second nozzle is positioned within the rear panel and is oriented to direct a second jet of fluid in an upward direction toward the access aperture. The first and second jets of fluid define a three-dimensional pattern of steam that engages the load of articles from above and below, respectively.

(61) According to yet another aspect, the operating pattern is based upon at least one of a fabric type that is being processed in the processing chamber and an amount of articles that are being processed in the processing chamber.

(62) According to another aspect of the present disclosure, the operating pattern includes activations and deactivations of the first nozzle and the second nozzle.

(63) According to another aspect, the plurality of sensors includes a conductivity sensor for measuring a humidity of process air within the processing chamber, a rear temperature probe and a front temperature probe. The rear and front temperature probes measure a temperature difference between the process air entering the processing chamber and the process air leaving the processing chamber. The humidity and the temperature difference cooperate to define the operating pattern.

(64) It will be understood by one having ordinary skill in the art that construction of the described disclosure and other components is not limited to any specific material. Other exemplary embodiments of the disclosure disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

(65) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

(66) It is also important to note that the construction and arrangement of the elements of the disclosure as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(67) It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

Claims

1. A laundry appliance comprising: a cabinet having an access aperture; a blower that delivers process air through an airflow path; a rotating drum positioned within the cabinet, the rotating drum defining a processing chamber, wherein the processing chamber is part of the airflow path, wherein the access aperture provides selective access to the processing chamber; a steam generating system that disposes steam into the processing chamber; and a plurality of fluid nozzles that direct the steam into the processing chamber according to an operating pattern, wherein the plurality of fluid nozzles includes a first nozzle that is positioned within an upper portion of the processing chamber and a second nozzle that is positioned in a lower portion of the processing chamber, wherein the first nozzle is positioned above the access aperture and is oriented to direct a first jet of fluid in a downward direction toward a rear panel of the processing chamber, and wherein the second nozzle is positioned within the rear panel and is oriented to direct a second jet of fluid in an upward direction toward the access aperture.

2. The laundry appliance of claim 1, wherein the operating pattern is based upon a fabric type that is being processed in the processing chamber.

3. The laundry appliance of claim 1, wherein the operating pattern is based upon an amount of articles that are being processed in the processing chamber.
4. The laundry appliance of claim 1, wherein the operating pattern includes activations and deactivations of the plurality of fluid nozzles.
5. The laundry appliance of claim 1, further comprising a steam generator that generates steam having a temperature that is similar to an air temperature of the process air within the processing chamber, and wherein the steam generator delivers fluid to the first nozzle and the second nozzle according to the operating pattern.
6. The laundry appliance of claim 5, wherein the operating pattern includes selective activation of valves for the first nozzle and the second nozzle, respectively, wherein the steam generator includes the first nozzle and the second nozzle.
7. The laundry appliance of claim 6, further comprising: a conductivity sensor in communication with the steam generator and the valves, wherein the conductivity sensor measures a humidity level within the processing chamber, and wherein the operating pattern and the selective activation of the valves are operated at least partially according to the humidity level within the processing chamber as measured by the conductivity sensor.
8. The laundry appliance of claim 6, further comprising: a rear temperature probe positioned within the airflow path and proximate the rear panel of the processing chamber, the rear temperature probe measuring an air temperature of the process air entering the processing chamber; and a front temperature probe positioned within the airflow path and proximate a front panel of the processing chamber, the front temperature probe measuring the air temperature of the process air exiting the processing chamber wherein; the rear and front temperature probes measure a temperature change of the process air; and the operating pattern and the selective activation of the valves are operated at least partially according to the temperature change of the process air.
9. A laundry appliance comprising: a cabinet having an access aperture defined within a front panel of the cabinet; a drum that rotates within the cabinet, wherein the access aperture provides selective access to a processing chamber defined within the drum; a steam generating system that delivers steam to the processing chamber; a first nozzle that is coupled to the front panel and above the access aperture; and a second nozzle that is positioned within a rear panel of the processing chamber; wherein the first nozzle is positioned within an upper portion of the processing chamber and is oriented to direct a first jet of fluid in a downward direction toward a rear panel of the processing chamber; the second nozzle is positioned within a lower portion of the processing chamber and is oriented to direct the steam in an upward direction toward the access aperture; a steam generator delivers steam to the first nozzle and the second nozzle according to an operating pattern, wherein the operating pattern is determined by a plurality of sensors that are positioned in communication with the processing chamber; and the steam generator generates steam in the absence of a dedicated heating element.
10. The laundry appliance of claim 9, wherein the operating pattern includes selective operation of the first and second nozzles.
11. The laundry appliance of claim 9, wherein the steam generating system includes a steam generator that generates steam having a temperature that is similar to an air temperature of process air within the processing chamber, and wherein the steam generator delivers steam to the first nozzle and the second nozzle according to the operating pattern.
12. The laundry appliance of claim 11, wherein the steam generating system includes valves that are positioned between the steam generator and the first and second nozzles, wherein the operating pattern includes selective activation of the valves for the first nozzle and the second nozzle, respectively.
13. The laundry appliance of claim 12, further comprising: a conductivity sensor in communication with the steam generator and the valves, wherein the conductivity sensor is disposed on a surface of the drum and measures a humidity level of process air within the processing chamber, and wherein

the operating pattern and the selective activation of the valves are operated at least partially according to the humidity level, as measured by the conductivity sensor.

14. The laundry appliance of claim 9, wherein the operating pattern is based upon a fabric type that is being processed in the processing chamber.

15. The laundry appliance of claim 9, wherein the operating pattern is based upon an amount of articles that are being processed in the processing chamber.

16. The laundry appliance of claim 9, wherein the operating pattern includes a pattern of selective activations and deactivations of the first nozzle and the second nozzle.

17. A laundry appliance comprising: a cabinet having an access aperture; a drum that rotates within the cabinet to process a load of articles, wherein the access aperture provides selective access to a processing chamber defined within the drum; a steam generating system that delivers steam to the processing chamber; a first nozzle that is coupled to a front panel and above the access aperture; and a second nozzle that is positioned within a rear panel of the processing chamber; wherein the steam generating system delivers steam to the first nozzle and the second nozzle according to an operating pattern, wherein the operating pattern is determined by a plurality of sensors that are positioned in communication with the processing chamber; wherein the first nozzle is oriented to direct a first jet of fluid in a downward direction toward a rear panel of the processing chamber and the second nozzle is positioned within the rear panel and is oriented to direct a second jet of fluid in an upward direction toward the access aperture; and wherein the first and second jets of fluid define a three-dimensional pattern of steam that engages the load of articles from above and below, respectively.

18. The laundry appliance of claim 17, wherein the operating pattern is based upon at least one of a fabric type that is being processed in the processing chamber and an amount of articles that are being processed in the processing chamber.

19. The laundry appliance of claim 17, wherein the operating pattern includes activations and deactivations of the first nozzle and the second nozzle.

20. The laundry appliance of claim 17, wherein the plurality of sensors includes a conductivity sensor for measuring a humidity of process air within the processing chamber, a rear temperature probe and a front temperature probe, wherein the rear and front temperature probes measure a temperature difference between the process air entering the processing chamber and the process air leaving the processing chamber, wherein the humidity and the temperature difference cooperate to define the operating pattern.
